PageRank algorithm:
The 25 billion dollar eigenvector

The PageRank algorithm is one of many algorithms that Google uses to determine the relative importance of a webpage. In 1998, Brin and Page posed the original formulation and subsequent solution of the PageRank problem in the Markov chain realm. Google owes its success partly to its PageRank algorithm, which enables ranking the pages and thereby presenting to the user the most relevant and helpful pages first. Here, we will study a simplified version of this algorithm.

The PageRank algorithm models the internet with a directed graph. Each webpage is a node, and there is an edge from node i to node j if page i links to page j. Let In(i) be the websites linking to page i and let Out(i) be the websites that page i links to.

The PageRank algorithm ranks pages by how many other pages link to them. A link from a more important page counts more than one from a less important page. For example, in Figure (1) we would expect node 0 to have a very high rank because every other node links to it. Consequently, we would expect node 7 to have a fairly high rank because node 0 links to it, even though node 0 is the only node to do so.

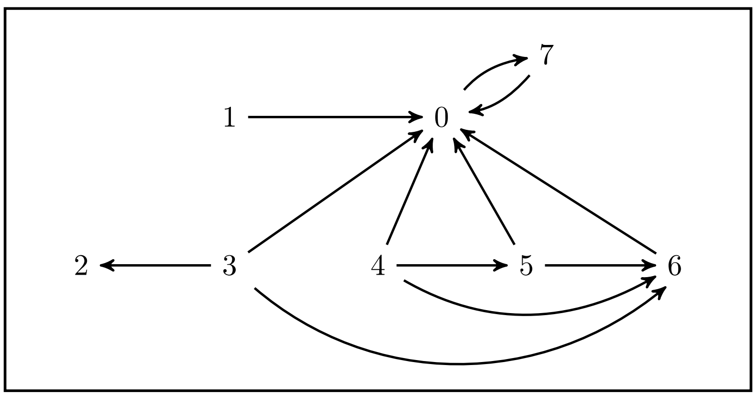


Figure (1): This directed graph describes the links between 8 webpages.

The PageRank algorithm assumes that a surfer chooses a starting webpage randomly. Then, if the surfer is at page i, they randomly select a page from Out(i) to visit next. This means that the surfer's chance of being on page i at time t is determined by where they were at time $t - 1$.

For example: $Out(2) = \{\}$, $out(4) = \{0,5,6\}$, $In(2) = \{3\}$, $In(4) = \{\}$.

Also, we will use the notation $|out(i)|$ to represent the number of webpages that link to page i. For example: $|Out(4)| = 3$.

Question 1: A surfer starts by choosing a webpage randomly. Suppose the internet has N webpages, and let $p_i(t)$ be the likelihood that the surfer is on page i at time t. What are the probabilities $p_i(0)$ and $p_i(t+1)$?

$$P_i(0) = \frac{1}{N}$$

$$P_i(t+1) = \sum_{j \in In(i)} \frac{P_j(t)}{|Out(j)|}$$

Sum over the in-linking pages

number of out-links of the linking page j

Key ideas: (I) A page is highly ranked if it is cited often by other pages. The importance of page i increases if it is linked to from page j. $(j \in Out(i))$

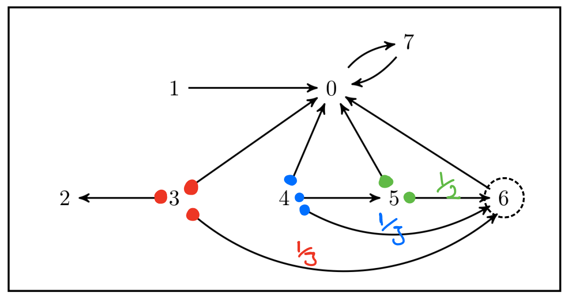
(II) Also, the increment in the importance of page i should depend on the importance of the linking page P_j;

The increment in the importance of page i should be inversely proportional to the number of out-links from page j which is denoted by $|Out(j)|$.

Question 2: Find $p_6(t+1)$.

$|Out(3)| = 3 \quad |Out(4)| = 3 \quad |Out(5)| = 2$

$$P_6(t+1) = \frac{P_3(t)}{3} + \frac{P_4(t)}{3} + \frac{P_5(t)}{2}$$



Question 3: For the graph in figure (1), find a matrix which will represent the evolution of probabilities from time t to time $t+1$.

$$\underbrace{\begin{bmatrix} 0 & 1 & 0 & 1/3 & 1/3 & 1/2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{H_0} \underbrace{\begin{bmatrix} P_0(t) \\ P_1(t) \\ \vdots \\ P_5(t) \\ P_6(t) \\ P_7(t) \end{bmatrix}}_{\vec{P}(t)} = \underbrace{\begin{bmatrix} P_0(t+1) \\ \vdots \\ \\ P_6(t+1) \\ P_7(t+1) \end{bmatrix}}_{\vec{P}(t+1)}$$

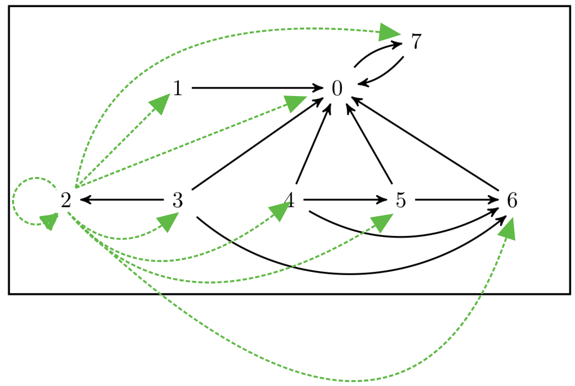
For example: $P_6(t+1) = \frac{P_3(t)}{3} + \frac{P_4(t)}{3} + \frac{P_5(t)}{2}$

Pages with No Outbound Links

On the Internet many pages are *dangling or sink links*, without any outbound links. What happens to the random surfer who visits such a page? The simplest thing is to imagine is that the surfer chooses the next page entirely at random.

Question 4: Modify the directed graph of Figure (1) and the probabilities $p_i(t+1)$ assuming that surfers visiting a page with no outbound link will choose another page at random.

$$H = \begin{bmatrix} 0 & 1 & 1/8 & 1/3 & 1/3 & 1/2 & 1 & 1 \\ 0 & 0 & 1/8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/8 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/8 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 1/8 & 1/3 & 1/3 & 1/2 & 0 & 0 \\ 1 & 0 & 1/8 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



$$P_6(t+1) = \frac{P_3(t)}{3} + \frac{P_4(t)}{3} + \frac{P_5(t)}{2} + \underbrace{\frac{P_2(t)}{8}}_{\text{additional Term}}$$

H is column stochastic and will preserve probabilities.

Making the model more realistic (Adding boredom)

Previously we assumed that the current page must link to the next page. However, the model is more realistic if we assume that the surfer sometimes gets bored and randomly picks a new starting page.

Question 5: Denote the probability that a surfer stays interested at time step t by a constant α, called the *damping factor*. How would the probabilities $p_i(t+1)$ be modified?

$$P_i(t+1) = \alpha\left(\sum_{j\in In(i)} \frac{P_j(t)}{|Out(j)|}\right) + (1-\alpha)\,\frac{1}{N}$$

with probability $(1-\alpha)$ the surfer loses interest and will choose another page at random.

Question 6: Express the result of previous question as a matrix equation and show that it represents a Markov chain whose transition matrix is regular and stochastic.

$$\vec{p}(t+1) = \alpha H \vec{p}(t) + (1-\alpha)\begin{bmatrix} \frac{1}{N} \\ \frac{1}{N} \\ \vdots \\ \frac{1}{N} \end{bmatrix}$$

$$\vec{p}(t+1) = \alpha H \vec{p}(t) + (1-\alpha)\begin{bmatrix} \frac{1}{N} & \frac{1}{N} & \cdots & \frac{1}{N} \\ \frac{1}{N} & \frac{1}{N} & \cdots & \frac{1}{N} \\ \vdots & & & \\ \frac{1}{N} & \frac{1}{N} & \cdots & \frac{1}{N} \end{bmatrix} \vec{p}(t)$$

$$\vec{p}(t+1) = \alpha H \vec{p}(t) + \frac{(1-\alpha)}{N}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & & \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{N\times N} \vec{p}(t)$$

$$\vec{p}(t+1) = \left(\alpha H + \frac{(1-\alpha)}{N}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & & \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{N\times N}\right) \vec{p}(t)$$

Note that H is a stochastic matrix and

$$B = \frac{1}{N}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & & \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{N\times N}$$ is also a stochastic matrix.

Therefore, $\alpha H + (1-\alpha)B$ is also a stochastic matrix:

$$\sum_{i=1}^{N} \alpha H_{ij} + (1-\alpha) B_{ij} = \alpha \sum_{i=1}^{N} H_{ij} + (1-\alpha)\sum_{i=1}^{N} B_{ij}$$
$$= \alpha(1) + (1-\alpha)(1)$$
$$= 1$$

Also, it is obvious that $\alpha H + (1-\alpha)B$ is a regular matrix.

Question 7: Choose $\alpha = 0.85$ and find the steady state of the Markov chain derived in the previous question for the example shown in Figure (1).

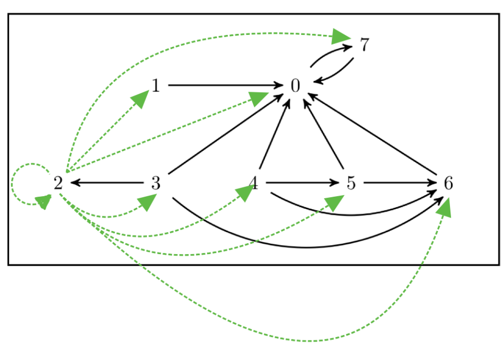
$$T = \alpha H + \frac{(1-\alpha)}{N}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & & \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{N\times N}$$

We calculate the eigenstate corresponding to eigenvalue one:

As expected page 0 has the highest rank.

$$\begin{bmatrix} 0.4387 \\ 0.0217 \\ 0.0279 \\ 0.0217 \\ 0.0217 \\ 0.0279 \\ 0.0459 \\ 0.3946 \end{bmatrix}$$

Page 7 has a higher rank than 6, because it has been linked to by the highest rank page (page 0).



The above probability vector provides a comparison between the importance of the webpages.